



João Ventura

Master in Distributed and Parallel

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Profile

First-year Masters student in Distributed and Parallel Systems at NOVA FCT. Passionate about servers, automation, cloud computing, and parallelization, with a strong focus on backend development and building RESTful systems. Seeking opportunities for a summer internship or a company partnership for my master's thesis project.

Academic Background

Bachelor's in Computer Science — NOVA FCT, Caparica (Expected 2026)

Gained comprehensive foundation in computer science fundamentals with particular interest in distributed systems, software engineering, and functional programming. Main coursework: Java and Python programming, Data Structures and Algorithms, Artificial Intelligence, and Machine Learning. Acted as tutor for programming and introductory AI courses.

Professional Experience

Land Management Application (Team Lead, 2025)

Led a team of four developers in designing and implementing a full-stack web and mobile platform for sustainable land management in collaboration with municipal partners. Built features for interactive GeoJSON-based mapping, field data collection with GPX tracking, real-time operation monitoring, photo and observation documentation, and offline mobile functionality. Developed secure user authentication with JWT and cookies, role-based access control, user management dashboards, and comprehensive reporting tools.

Home Server Setup & Exploration (2025)

Built a local server environment to self-host applications and explore system administration and networking. Currently using it to host personal projects and test distributed environments.

Distributed Server-Client Implementation (2025)

Designed and implemented a distributed server-client system focusing on replication, data consistency, and availability. Implemented fault tolerance and synchronization mechanisms using Zookeeper and Kafka.

Programming Tutor — TreeTree2 (2021)

Tutored young people in Python over 2 months of weekend sessions, helping students conceptualize their progress and grasp Python basics effectively.

Licenses and Certification

Programming Certificates — TreeTree2 (2020-2021)

Completed intermediate courses covering Python and C++ fundamentals, algorithm problem-solving, introductory AI model development, and game development with 2D platformer projects.

Technical Skills

Programming Languages: Java, Python, C, JavaScript, Shell, Dart

Frameworks & Tools: Clusters, Arduino, Spring Boot, Docker, Apache

Databases: SQL, PostgreSQL, Firebase, AzureDB

Distributed Systems: Zookeeper, Apache Kafka, Paxos, Raft

Web & Cloud: Google App Engine, Microsoft Azure

Version Control: Git, GitHub

Development: Web/ Mobile App Development, RESTful API Design

Other: Self-hosted portfolio website, Server Administration

Projects

Portfolio Website — jhonceda.duckdns.org

Self-hosted portfolio built with HTML, CSS, and JavaScript. Plans to integrate database functionality and RESTful APIs.

Awards & Recognition

Honor Roll (2021): Awarded for exceptional dedication to supporting classmates; the only student recognized with this distinction that year.

Languages

Portuguese: Native **English:** Advanced (C2) **Spanish:** Intermediate (B1) **French:** Beginner A2

Hobbies

Cooking, music (listening, composing, playing), technology tinkering, and creating art.